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Can you trust an AI agent with FDA reportability decisions?

We tested whether AI agents give the same, correct answer when a medical device complaint must be judged for FDA reporting.

How the test works

📋 40 real complaints (FDA reports + consumer reviews) →

🔁 Each written 5 ways, asked 3 times →

⚙ 6 agent setups compared

Meet the 6 agents

Haiku · simple prompt
Decides without a rulebook.

Haiku · simple + rulebook
Can consult FDA rules.

Haiku · step-by-step
Decides without a rulebook.

Haiku · step-by-step + rulebook
Can consult FDA rules.

Sonnet · simple + rulebook
Can consult FDA rules.

Opus · simple + rulebook
Can consult FDA rules.

Rulebook = a regulatory lookup tool. Haiku, Sonnet and Opus are different sizes of the same model family.

› What the measures mean

Device: All | Pulse oximetry | Glucose meter

Case difficulty: All | Clear | Borderline

Source: All | FDA reports | Consumer reviews

Wording: All | W1 | W2 | W3 | W4 | W5

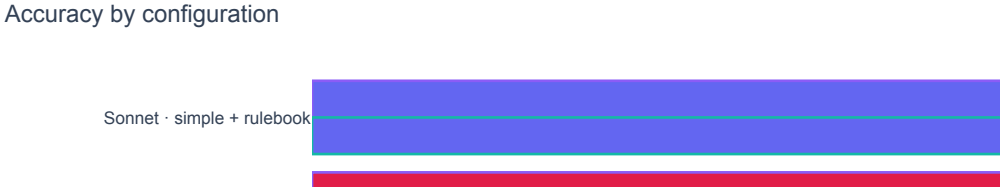
n = 3,600 tests

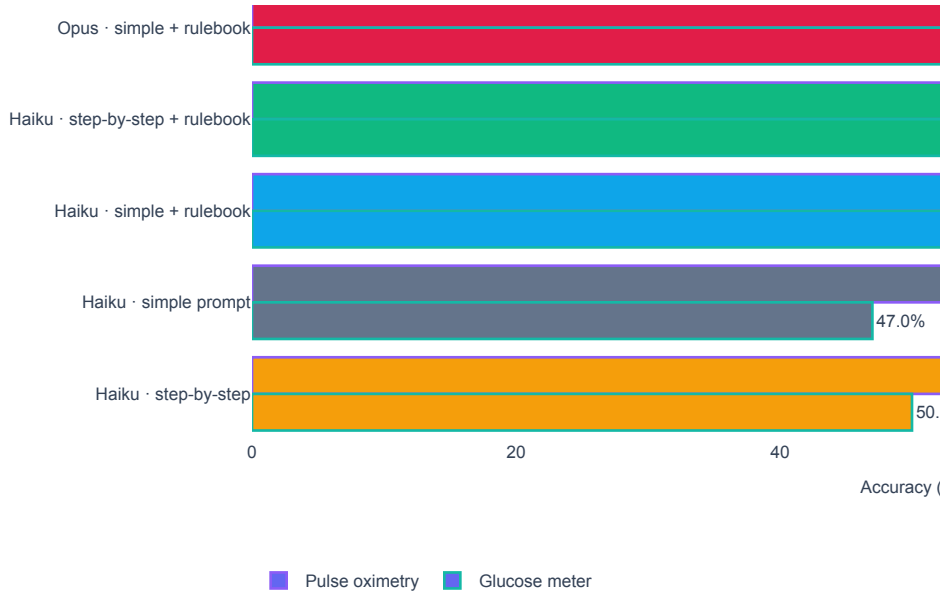
Tests scored	Best accuracy	Best consistency	Under-reporting rate
3,600	80.7%	77.5%	0.0%

Compare configurations

Metric: Accuracy

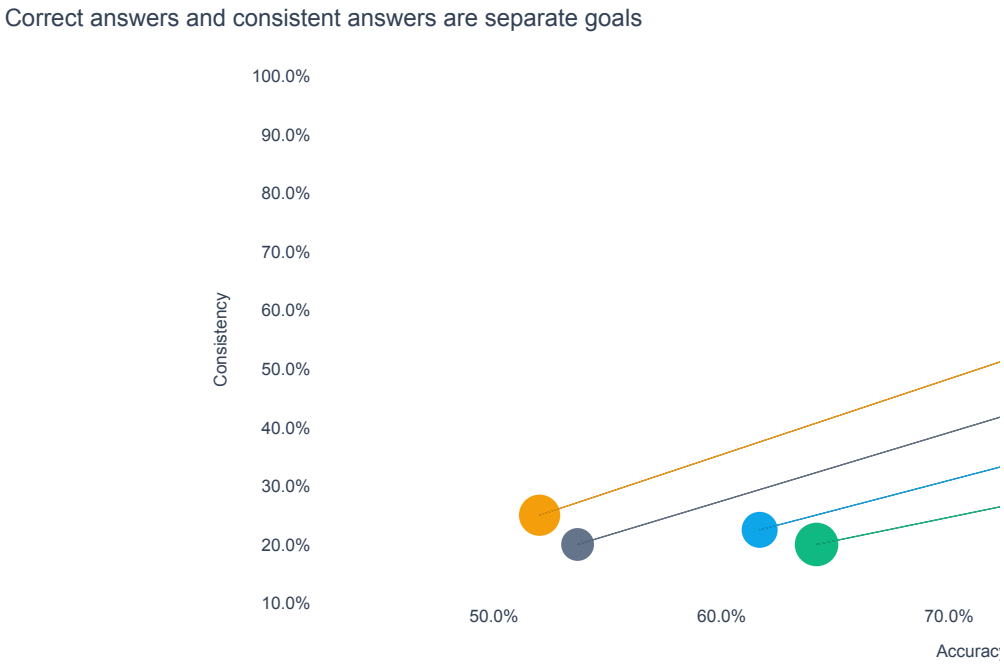
Group by: All six | Haiku prompt designs | Model sizes with a rulebook





Click a bar to inspect outcomes and the five hardest cases. Outlines identify devices.

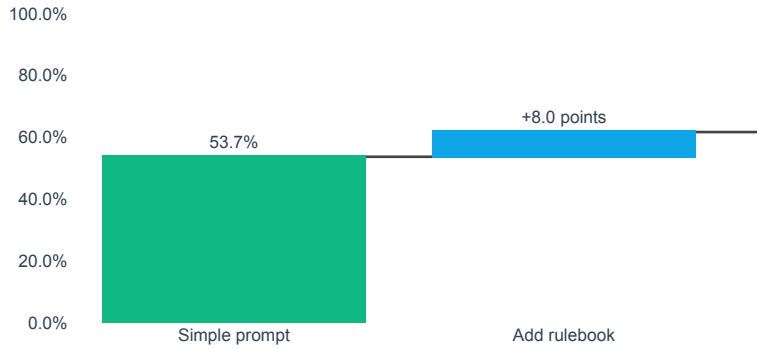
How often are answers correct and repeatable?



Bubble size = tokens. H1–H4 identify the four Haiku cards in order; hover for full names. The target is illustrative.

What do the prompt and rulebook change?

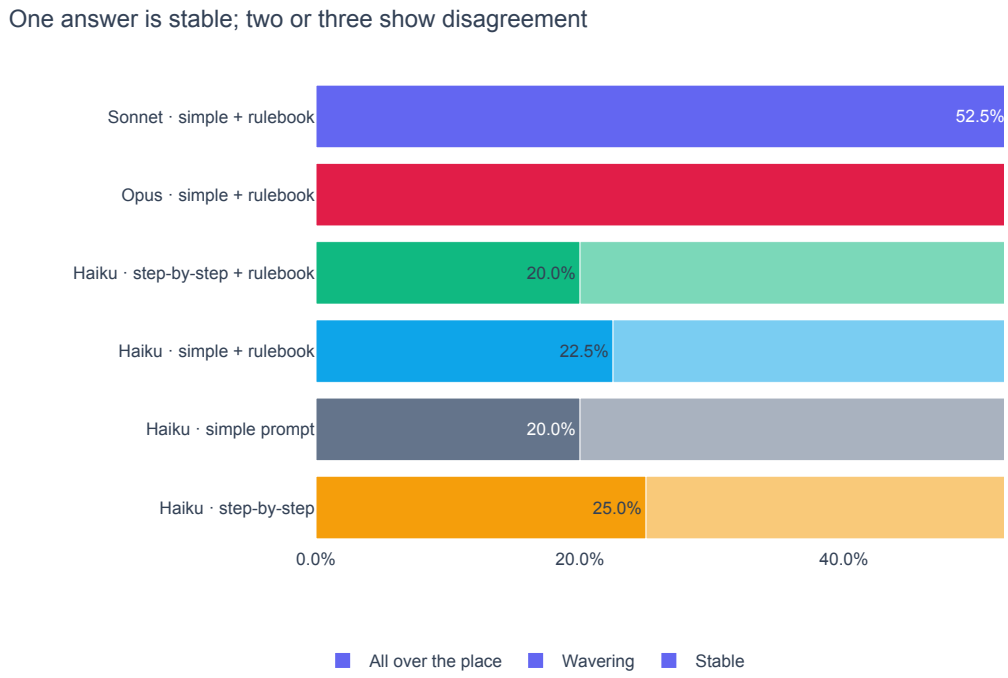
Haiku accuracy changes along one design path



	No rulebook	Rulebook
Simple	53.7%	61.7%
Step-by-step	52.0%	64.2%

600 matching tests per design; the table shows accuracy for all four Haiku agents.

How often does an agent change its answer?



Stable = one answer; wavering = two; all over the place = three. Invalid responses are excluded; partial cases use available attempts.

What kind of mistakes?

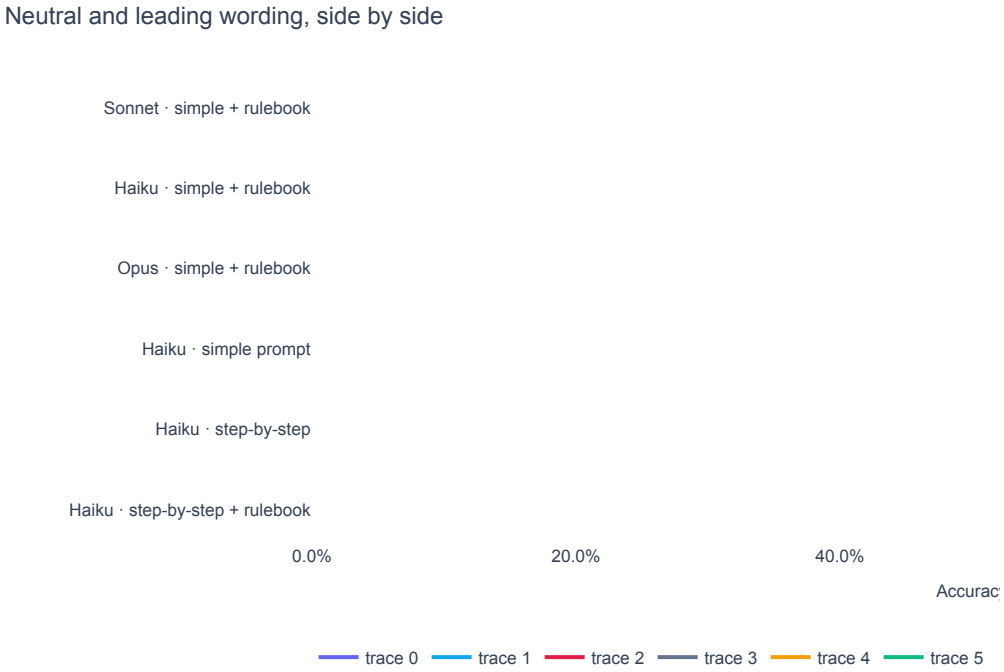
Agent

Opus · simple + rulebook



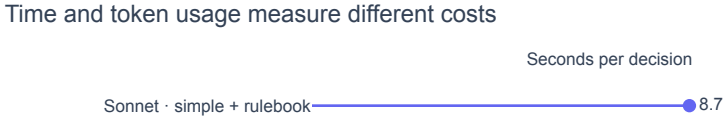
Each bar totals 100% of tests with that expected answer.

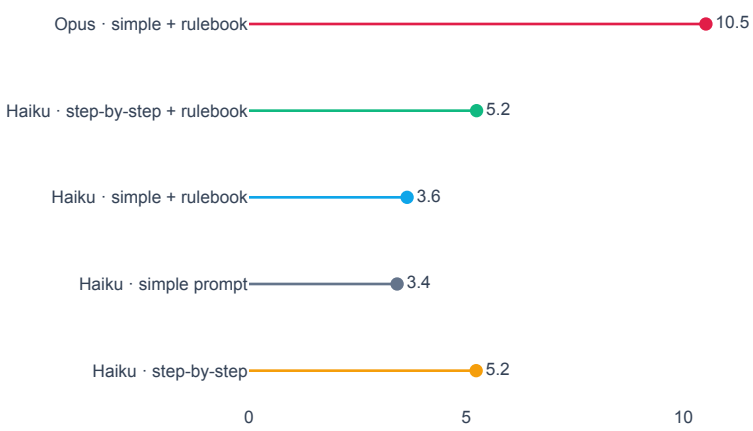
Do leading questions change accuracy?



Circle = neutral; diamond = leading. Cave-in is neutral accuracy minus leading accuracy, in percentage points.

How much time and usage does each answer cost?





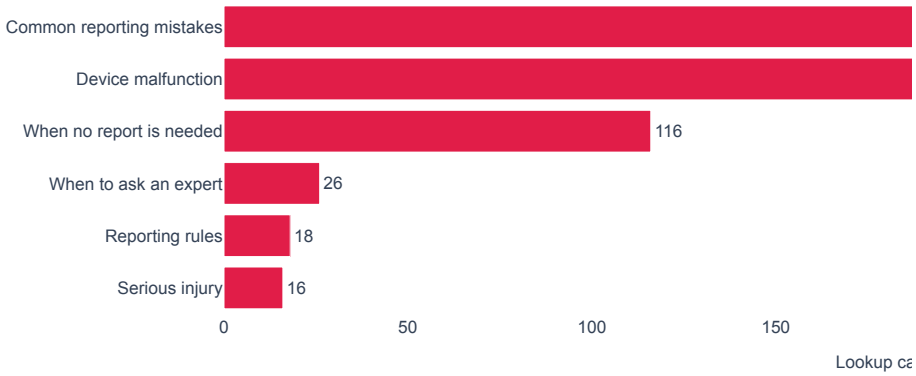
Averages include all turns in the latest attempt. Tokens measure usage, not price.

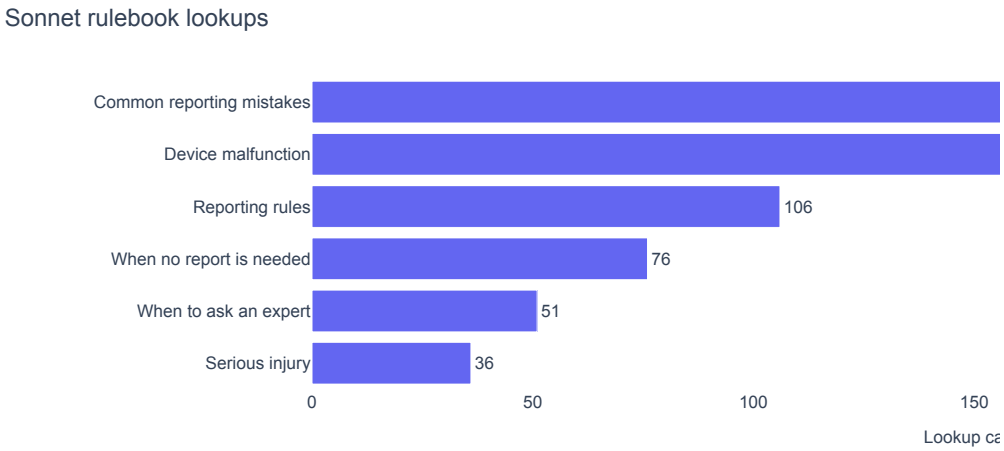
Which rulebook topics did the models consult?

Haiku rulebook lookups



Opus rulebook lookups





Calls include repeated topics; agents without recorded lookups are omitted.

Details

What this means for the quality team

- 🎯 Sonnet · simple + rulebook matches the expert on **80.7%** of tests.
- 🔁 Opus · simple + rulebook keeps the same answer on **77.5%** of cases.
- ⚠️ Under-reporting is **0.0%** among reportable tests; other mistakes still need review.

These takeaways use the full experiment, regardless of the filters above.

Use as AI-assisted triage with human review, not unsupervised.